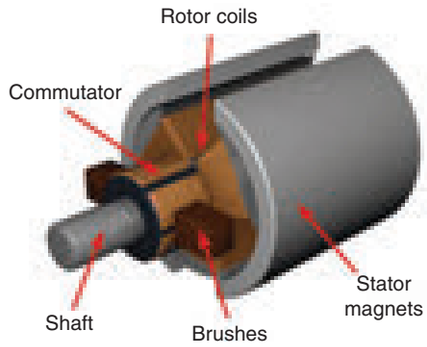


Fig. 5.3 DC motor

5.2.3 AC drive

AC drive is a type of electric machine, and it is shown in figure 5.4. It changes the electrical energy into mechanical energy. The inner appearance of the AC Motor is shown in figure 5.5. AC motors are used in electric fan, electric pump, air compressor and automobile industries.

AC motors has two parts.

1. Stator.
2. Rotor.



Fig. 5.4 AC motor

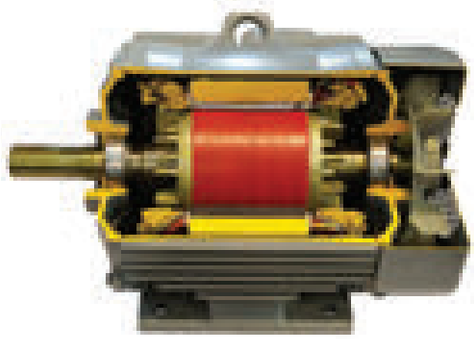


Fig. 5.5 Internal structure of AC motor

5.2.4 Comparison of DC and AC drives

Sl. No.	DC drive	AC drive
1	Energy and method of control is simple	Energy and method of control is difficult
2	More maintenance is required.	Less maintenance is required.
3	Commutator is required	Commutator is not required
4	Rapid speed control	Gradual speed control
5	Speed and design ratings are less	Speed and design ratings are high

5.2.5 Advantages of electric drives

i. It has flexible control characteristics.
ii. Drives with fixed and dynamic characteristics changed according to load requirements.
iii. Computers can be used to control the drive movements automatically by closed loop operation.
iv. Available with wide range of torque, speed and power.
v. It is applicable to function in the state such as explosion and radiation.
vi. It can start immediately even at full load of the drive.
vii. To control the speed and also simplest way to start and stop the drive.

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